

mTOR is a key modulator of ageing and age-related disease

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Many experts in the biology of ageing believe that pharmacological interventions to slow ageing are a matter of ‘when’ rather than ‘if’. A leading target for such interventions is the nutrient response pathway defined by the mechanistic target of rapamycin (mTOR). Inhibition of this pathway extends lifespan in model organisms and confers protection against a growing list of age-related pathologies. Characterized inhibitors of this pathway are already clinically approved, and others are under development. Although adverse side effects currently preclude use in otherwise healthy individuals, drugs that target the mTOR pathway could one day become widely used to slow ageing and reduce age-related pathologies in humans.